



Important Formulas: Fractions

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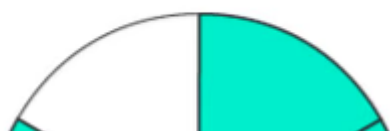
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1. **Fraction:** A fraction represents a **part of a whole**. When a whole object (for example, a pizza) is divided into equal parts, each part is a fraction of that whole.
2. **Fractional unit:** A fractional unit is **one equal part** of the whole after it has been divided.
Example: In the fraction **$\frac{1}{5}$** , the fractional unit is one part out of five equal parts of the whole.

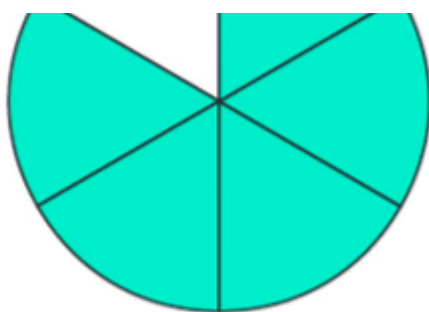
Reading fractions

Numerator: The number on top of the fraction. It tells how many parts of the whole we have.

Denominator: The number at the bottom of the fraction. It tells into how many equal parts the whole is divided into.



Numerator
(number of parts we have)



$$\frac{5}{6}$$

Denominator
(total parts in whole)

Number line

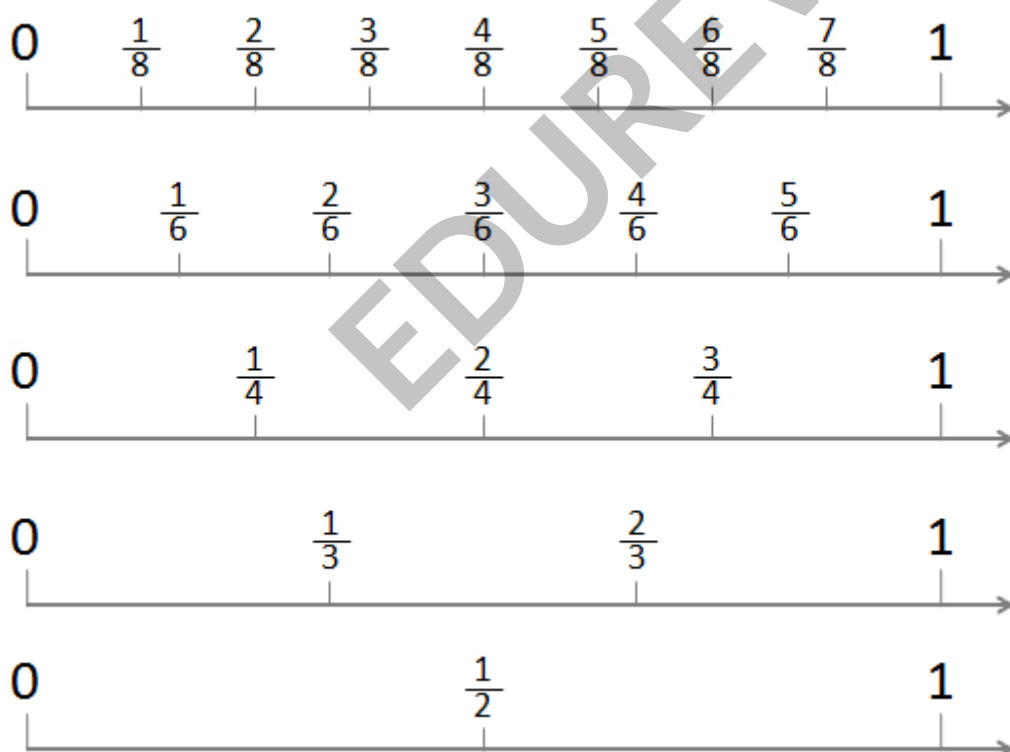
A number line shows where fractions lie between whole numbers. The segment between 0 and 1 is divided into equal parts according to the denominator.

Every fraction corresponds to a point on the number line.

Example,

$\frac{1}{2}$ is halfway between 0 and 1

$\frac{3}{4}$ is three of the four equal parts from 0 towards 1.



Mixed fractions

A mixed fraction (or mixed number) is a combination of a whole number and a proper fraction.

Example: $2\frac{1}{7}$ means 2 whole units and 1 part out of 7 equal parts of another unit.

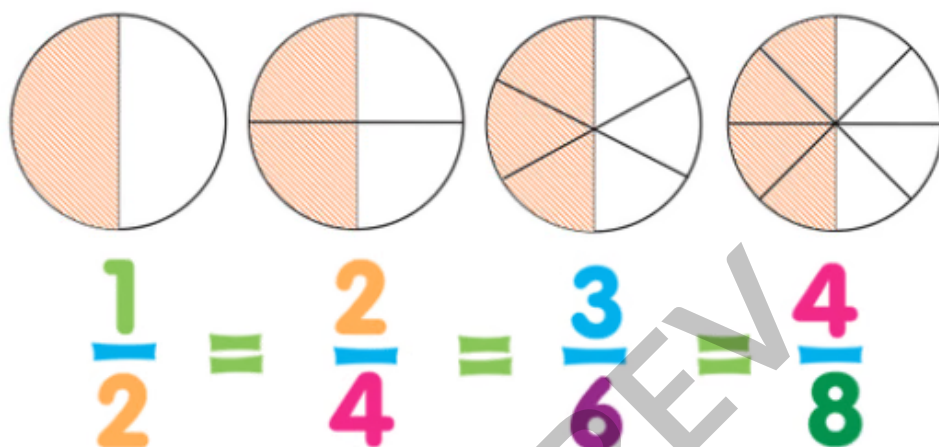
Whole number $2 \frac{1}{7}$ Fraction

Equivalent fractions

Equivalent fractions are different fractions that represent the same part of a whole.

Method: Multiply or divide both numerator and denominator by the same non-zero number to get an equivalent fraction.

Example: $\frac{1}{2}$ is equivalent to $\frac{2}{4}$ because $1 \times 2 = 2$ and $2 \times 2 = 4$.



MULTIPLE CHOICE QUESTION

Try yourself: Which option gives an equivalent fraction of $\frac{13}{25}$?

A $\frac{65}{50}$

B $\frac{26}{75}$

C $\frac{156}{300}$

D $\frac{103}{205}$

[View Solution](#)

Simplest form

A fraction is in its simplest form when the numerator and denominator have no common factor other than 1.

Method: Divide the numerator and the denominator by their greatest common divisor (GCD) to simplify the fraction.

Example: To simplify $\frac{6}{9}$, divide numerator and denominator by 3 to get $\frac{2}{3}$.

Comparing fractions

Method 1: Convert the fractions to a common denominator and compare the numerators.

Method 2: Use a number line to see which fraction is located further to the right (that fraction is greater).

Example: Compare $\frac{4}{5}$ and $\frac{7}{9}$.

(i) Find a common denominator: The denominators are 5 and 9. The least common multiple (LCM) of 5 and 9 is 45.

(ii) Convert each fraction to denominator 45:

$$\frac{4}{5} = \frac{(4 \times 9)}{(5 \times 9)} = \frac{36}{45}$$

$$\frac{7}{9} = \frac{(7 \times 5)}{(9 \times 5)} = \frac{35}{45}$$

$$\begin{aligned}\frac{4}{5} &= \frac{4 \times 9}{5 \times 9} = \frac{36}{45} \\ \frac{7}{9} &= \frac{7 \times 5}{9 \times 5} = \frac{35}{45}\end{aligned}$$

Since $\frac{36}{45} > \frac{35}{45}$, we conclude that $\frac{4}{5} > \frac{7}{9}$.

Addition of fractions

If denominators are the same (**like fractions**), add the numerators and keep the denominator

the same. Simplify the result if possible.

If denominators are different (**unlike fractions**), first find the least common denominator (LCD), convert each fraction to an equivalent fraction with the LCD, then add the numerators.

Example: $\frac{1}{4} + \frac{1}{4}$

Sol:

Both fractions have the same denominator (4).

Add the numerators: $1 + 1 = 2$.

Write the result over the common denominator: $\frac{2}{4}$.

Simplify $\frac{2}{4}$ by dividing numerator and denominator by 2 to get $\frac{1}{2}$.

The diagram illustrates two methods for adding fractions, separated by a vertical line. On the left, under a pink header 'Same Denominators', the fractions $\frac{1}{5} + \frac{3}{5}$ are shown. A pink arrow points down to the result $\frac{1+3}{5} = \frac{4}{5}$, where the final fraction is circled in green. On the right, under a red header 'Different Denominators', the fractions $\frac{1}{2} + \frac{3}{7}$ are shown. Above the first fraction, 7×1 is written above the numerator and 7×2 above the denominator. Above the second fraction, 3×2 is written above the numerator and 7×2 above the denominator. A red arrow points down to the result $\frac{7+6}{14} = \frac{13}{14}$, where the final fraction is circled in green.

Example: $\frac{5}{8} + \frac{11}{24}$

The given fractions are unlike fractions,
so we first find the LCM of their denominators.

2	8, 24
2	4, 12
2	2, 6
3	1, 3
	1, 1

LCM of 8 and 24 = $2 \times 2 \times 2 \times 3 = 24$

Now, we convert the fractions into like fractions with denominator 24.

$$\frac{5 \times 3}{8 \times 3} = \frac{15}{24} \text{ and } \frac{11}{24}$$

$$\frac{15}{24} + \frac{11}{24} = \frac{26}{24}$$

Subtraction of fractions

If denominators are the same, subtract the numerators and keep the denominator the same.
Simplify the result if possible.

If denominators are different, first find the least common denominator (LCD), convert each fraction to an equivalent fraction with the LCD, then subtract the numerators.

Example: For $\frac{3}{4} - \frac{2}{4}$, subtract numerators: $3 - 2 = 1$, so the answer is $\frac{1}{4}$.

Same Denominators	Different Denominators
$\frac{3}{5} - \frac{1}{5}$ ↓ $\frac{3-1}{5} = \frac{2}{5}$	$\frac{7 \times 1}{7 \times 2} - \frac{3 \times 2}{7 \times 2}$ ↓ $\frac{7-6}{14} = \frac{1}{14}$

Example: Subtract $\frac{4}{7}$ from $\frac{8}{3}$.

Solution:

Identify the denominators: 3 and 7.

Find the LCM of 3 and 7:

$$\text{LCM}(3, 7) = 21.$$

Convert $\frac{8}{3}$ to an equivalent fraction with denominator 21:

$$\frac{8}{3} = \frac{(8 \times 7)}{(3 \times 7)} = \frac{56}{21}.$$

Convert $\frac{4}{7}$ to an equivalent fraction with denominator 21:

$$\frac{4}{7} = \frac{(4 \times 3)}{(7 \times 3)} = \frac{12}{21}.$$

Subtract the numerators over the common denominator:

$$\frac{56}{21} - \frac{12}{21}$$

$$= \frac{(56 - 12)}{21}.$$

Compute the difference in the numerator: $56 - 12 = 44$.

The result is $\frac{44}{21}$.

Answer: $\frac{8}{3} - \frac{4}{7} = \frac{44}{21}$.

Additional notes and tips

Always look for opportunities to simplify fractions early by cancelling common factors before performing addition or subtraction; this often makes arithmetic easier.

When converting fractions to a common denominator, using the LCM keeps the numbers smaller than multiplying denominators together in many cases.

Use a number line for intuitive understanding and quick comparison of simple fractions.

Remember the terms: numerator (top) and denominator (bottom). Practice converting between improper fractions and mixed numbers.